

Likun Xie

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Research Interests

Number theory: sieve methods and arithmetic over number fields and function fields; commutative algebra: homological methods and local cohomology.

Appointments

Max Planck Institute for Mathematics, Bonn, Germany

July 2025–present

Postdoctoral Fellow.

Education

University of Illinois Urbana–Champaign, United States

2020–2025

Ph.D. in Mathematics. Advisors: Alexandru Zaharescu and Bruce C. Berndt.

University of Warwick, United Kingdom

2018–2019

M.Sc. in Mathematics, Distinction. Dissertation: *Descent, Affine Representability and Classifying Space in $\mathbb{A}^1$ -Homotopy Theory*. Advisor: Marco Schlichting.

University of Manchester, United Kingdom

2016–2018

B.Sc. (Hons) in Mathematics and Physics, First Class. Dissertation: *Topos Foundation for Set Theory*. Advisor: Michael Prest.

Publications and Preprints

1. L. Xie, Connectedness in Codimension One and the Non- S_2 Locus, 2026. arXiv:2605.06617.
2. L. Xie, On a Conjecture of Erdős over Function Fields, 2025. arXiv:2510.17612. To appear in *Mathematische Zeitschrift*.
3. L. Xie, Almost Prime Orders of Elliptic Curves Modulo Prime Powers, 2025. arXiv:2504.18732.
4. K. Nath and L. Xie, Almost primes and primes of the form $p = 1 + m^2 + n^2$, 2025. arXiv:2501.16723. To appear in *Acta Arithmetica*.
5. B. C. Berndt and L. Xie, Identities for the product of two Dirichlet series satisfying Hecke-type functional equations, *Journal of Mathematical Analysis and Applications* 550(1), 129514.
6. L. Xie, On primes p such that $p - b$ has a large prime power factor and few other divisors, 2024. arXiv:2410.14133.
7. L. Xie, Negative Real Moments of Restricted Factors in Arithmetic Progression, *The Ramanujan Journal* 67, Article 41.
8. L. Xie, On an Instance of the Small Cohen–Macaulay Conjecture, *Journal of Pure and Applied Algebra* 228(3), 107491.
9. B. C. Berndt, L. Xie, and A. Zaharescu, Proofs of McIntosh’s Conjecture on Franel Integrals and Two Generalizations, *Advances in Mathematics* 423, 109041.

Awards

- Dr. Lois M. Lackner Mathematics Fellowship, University of Illinois Urbana–Champaign, 2023–2025.

Teaching and Mentoring

- Mentor, MPIM Internship Program, September 2025. Guided high-school and undergraduate interns on number theory projects related to sieve methods.
- Teaching Assistant, University of Illinois Urbana–Champaign: Calculus, Differential Equations, and Algebra, 2020–2023.
- Grader, University of Illinois Urbana–Champaign: Abstract Algebra, Honors Abstract Algebra, Commutative Algebra, Real Analysis, Complex Variables, and Differentiable Manifolds, 2020–2024.